

Monthly Intelligence Report

Edition 003 — The NSW–VIC Interconnector Corridor

Where demand surges are outrunning transmission capacity. edition · June 2026 · SSI v4.0.2

CONTENTS

A Executive Summary	E Asia-Pacific Context Card
B Cyber & Economic Exposure Monitor	F SSI Enhanced Neural Network
C Data Refresh & Changelog	G Looking Ahead
D Deep-Dive: The NSW–VIC Interconnector	

SECTION A

Executive Summary

SUBSTATIONS SCORED

8,500

Major + distribution across NEM & WA/NT

GRID LINES MAPPED

~9,200

~67,000 km transmission & distribution

MC SIMULATIONS

85.0 M

10,000 per substation

PUBLIC DATA SOURCES

20+

95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions. Its engine processes 146,000+ source data points per run, executes 85.0 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 1.78 billion arithmetic operations — producing 469,404 output data points with full uncertainty quantification across 8,500 substations and ~9,200 grid lines spanning ~67,000 km.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 20+ verified public sources (AEMO, AER, ABS, BOM, CSIRO, Geoscience Australia, CER), making the methodology fully auditable and reproducible. Initially covering Australia's NEM and isolated systems, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Shepparton SwStn 995

Victoria, Victoria · 33 kV

0.630

High

0.204

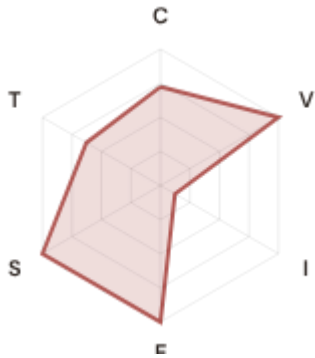
98th

R_FINAL

CLASSIFICATION

CI WIDTH

FLEET PERCENTILE



COMPONENTS

C Continuity: 0.725 / 0.30 (242%) ⬆

I Infrastructure: 0.123 / 0.25 (49%)

S Saturation: 1.254 / 0.20 (627%) ⬆

V Voltage: 2.681 / 0.10 (2681%) ⬆

E Economic: 1.311 / 0.10 (1311%) ⬆

T Transition: 0.628 / 0.05 (1256%) ⬆

MODIFIERS

R3 Consequence: 0.926 ⬇ dampens

R4 Graph Criticality: 0.781 ⬇ dampens

R6a Restoration: 0.922 ⬇ dampens

R6b Seismic: 1.018 ⬆ amplifies

R7 Digital Readiness: 1.006 → neutral

This month's spotlight — automatically selected for June 2026 — is **Shepparton SwStn 995** in Victoria, Victoria. With an R_final of 0.630 (High band, 98th fleet percentile), its risk profile is dominated by the V (Voltage) component at 2681% of its weight ceiling, followed by E (Economic) at 1311%. Modifiers collectively shift R_base by -32.2%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Australia's Energy Security Board and network operator strategic planning requires.

CYBER HIGH EXPOSURE

0
0.0% of fleet

BLIND SPOTS

711
High/Critical + R7 < median

AVG R7 MODIFIER

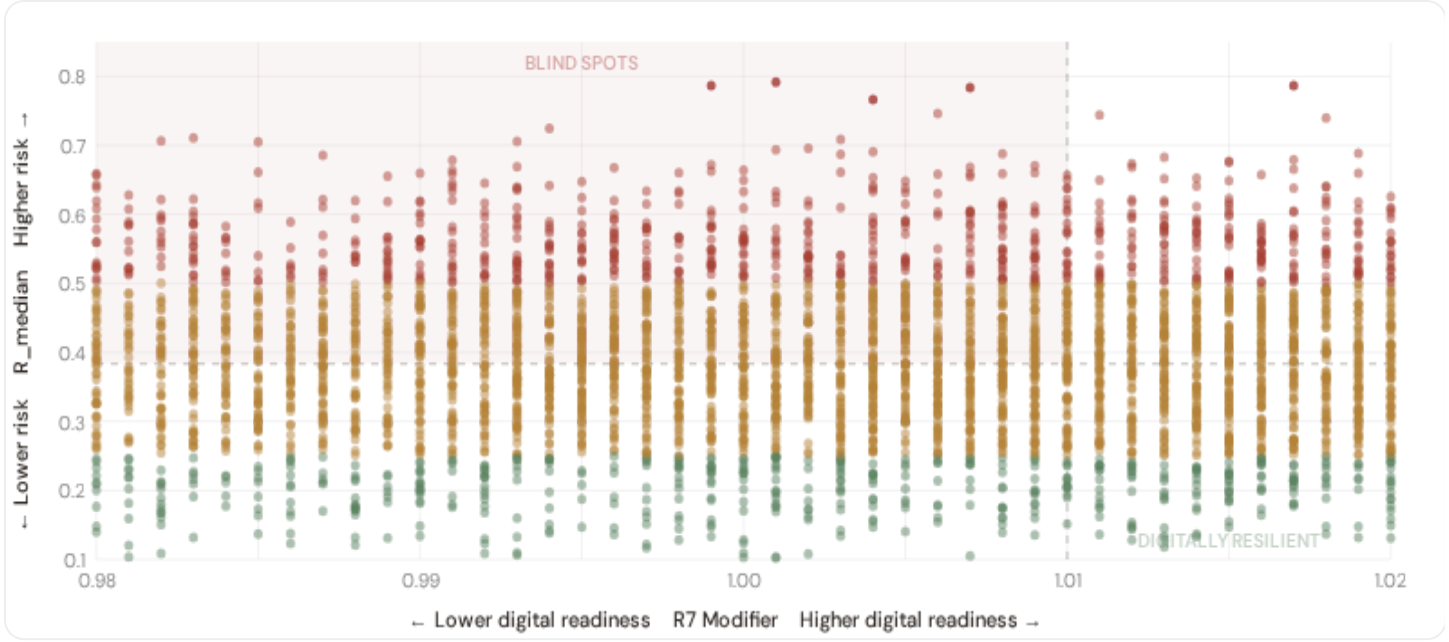
1.0101
Range [0.99, 1.05]

HIGH-CONSEQUENCE SUBS

2,126

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.

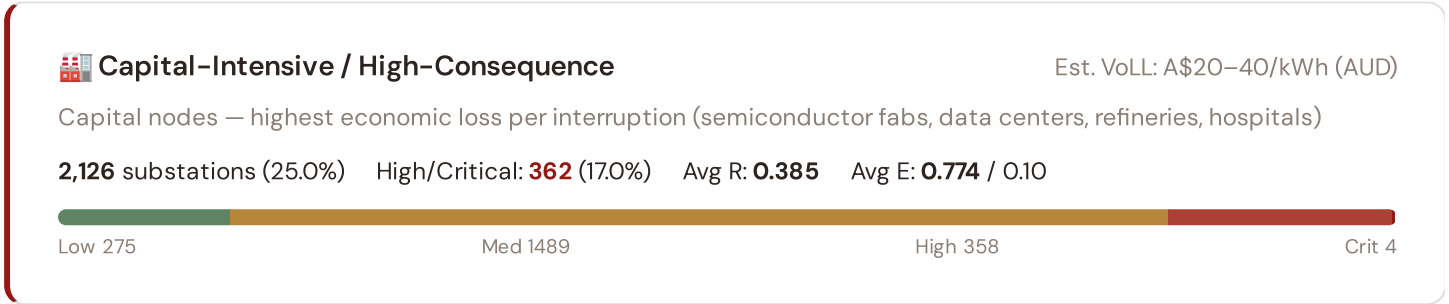


● Low ● Medium ● High ● Critical Dashed lines = fleet medians

The cyber-physical matrix identifies **711 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness ($R7 < 1.0100$). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **New South Wales (211), Queensland (173), Victoria (111)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.



Industrial / Medium–Large Enterprise

Est. VoLL: A\$10–20/kWh (AUD)

Industrial belt — significant continuity requirements (manufacturing, logistics hubs)

2,097 substations (24.7%) High/Critical: **336** (16.0%) Avg R: **0.381** Avg E: **0.795** / 0.10



Commercial / SME–Dense

Est. VoLL: A\$4–10/kWh (AUD)

Service economy + retail — moderate individual impact (commercial districts, mixed-use)

2,122 substations (25.0%) High/Critical: **387** (18.2%) Avg R: **0.387** Avg E: **0.845** / 0.10



Light / Agricultural / Rural

Est. VoLL: A\$1–4/kWh (AUD)

Sparse activity, agricultural land — lower VoLL (residential, agricultural)

2,155 substations (25.4%) High/Critical: **374** (17.4%) Avg R: **0.386** Avg E: **0.879** / 0.10

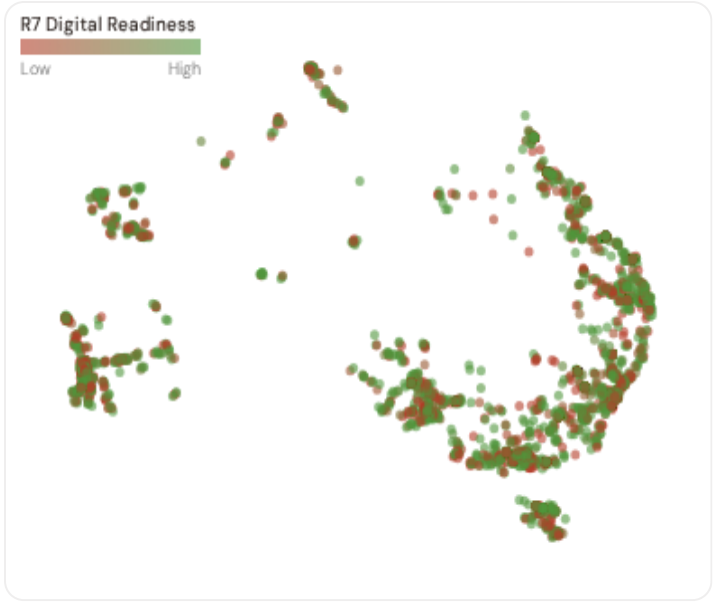


Value of Lost Load (VoLL) context: ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **362 substations** combine capital-intensive economic fabric ($R3 \geq 0.99$) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **New South Wales (669), Queensland (466), Victoria (424)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 8.8%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 State & SA4 Digital Readiness Ranking

SA4 regions ranked by average R7 modifier — lower values indicate weaker digital infrastructure (broadband penetration, smart meter coverage). Cross-referenced with average R_{median} to identify regions combining digital exposure with high grid stress.



WORST DIGITAL READINESS — TOP 15 REGION REGIONS			
Longer bar = higher R7 = better digital infrastructure.			
1	Northern Territory Δ high R		1.0082
2	Western Australia Δ high R		1.0087
3	Victoria		1.0096
4	Tasmania		1.0098
5	South Australia Δ high R		1.0104
6	Queensland Δ high R		1.0106
7	New South Wales		1.0107
8	Australian Capital Territory		1.0115

region-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Northern Territory** has the lowest average R7 (1.0082), while **Australian Capital Territory** leads at 1.0115 — a spread of 0.0033 across the R7 range. **2 region regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (June 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0101, median = 1.0100; 0 substations classified HIGH cyber-exposure; 711 blind-spot substations (High/Critical risk + below-median R7); 4 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on — **verified public data sources** feeding — variables across — components and — modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES

— variables

WEEKLY / MONTHLY

High-frequency feeds

QUARTERLY / ANNUAL

Regular refresh cycle

STATIC / DERIVED

Data Source Registry — June 2026

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

C.2 Fleet Score Snapshot

FLEET SIZE

8,500

substations scored

MEDIAN R

0.383

P5=0.188 · P95=0.585

HIGH + CRITICAL

1459

17.2% of fleet

HIGH CONFIDENCE

51%

4311 subs · CI/R < 0.50

BAND DISTRIBUTION



No primary data sources have been updated since the v4.0.2 launch baseline. All **8,500 substation scores remain unchanged** this edition. The fleet median R of 0.383 (mean 0.385) reflects a distribution skewed toward the lower bands, with 13.2% in Low and 69.6% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

0 changes tracked across PO, P1, and P2 releases

SECTION D — MONTHLY DEEP-DIVE

The NSW–VIC Interconnector Corridor: Where Demand Surges Outrun Transmission Capacity

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** NSW–VIC Interconnector corridor · **002** South Australian DER Saturation · **003** Queensland Tropical Infrastructure Stress · **004** Metropolitan–Remote economic impact disparity · **005** Coal closure transition risk (Hunter Valley) · **006** Infrastructure age vs smart meter rollout · **007** Stress–Resilience quadrant trends · **008** T-component forward projections · **009** SA4 Grid Health Cards · **010** Academic validation and peer feedback · **011** Asia–Pacific replication feasibility · **012** Year in review.

D.1 Why the NSW–VIC Corridor, Why Now

NSW SUBSTATIONS

120

45 EHV · 75 HV

HIGH BAND

43

35.8% of corridor

CORRIDOR MEDIAN R

0.436

Fleet median: 0.383

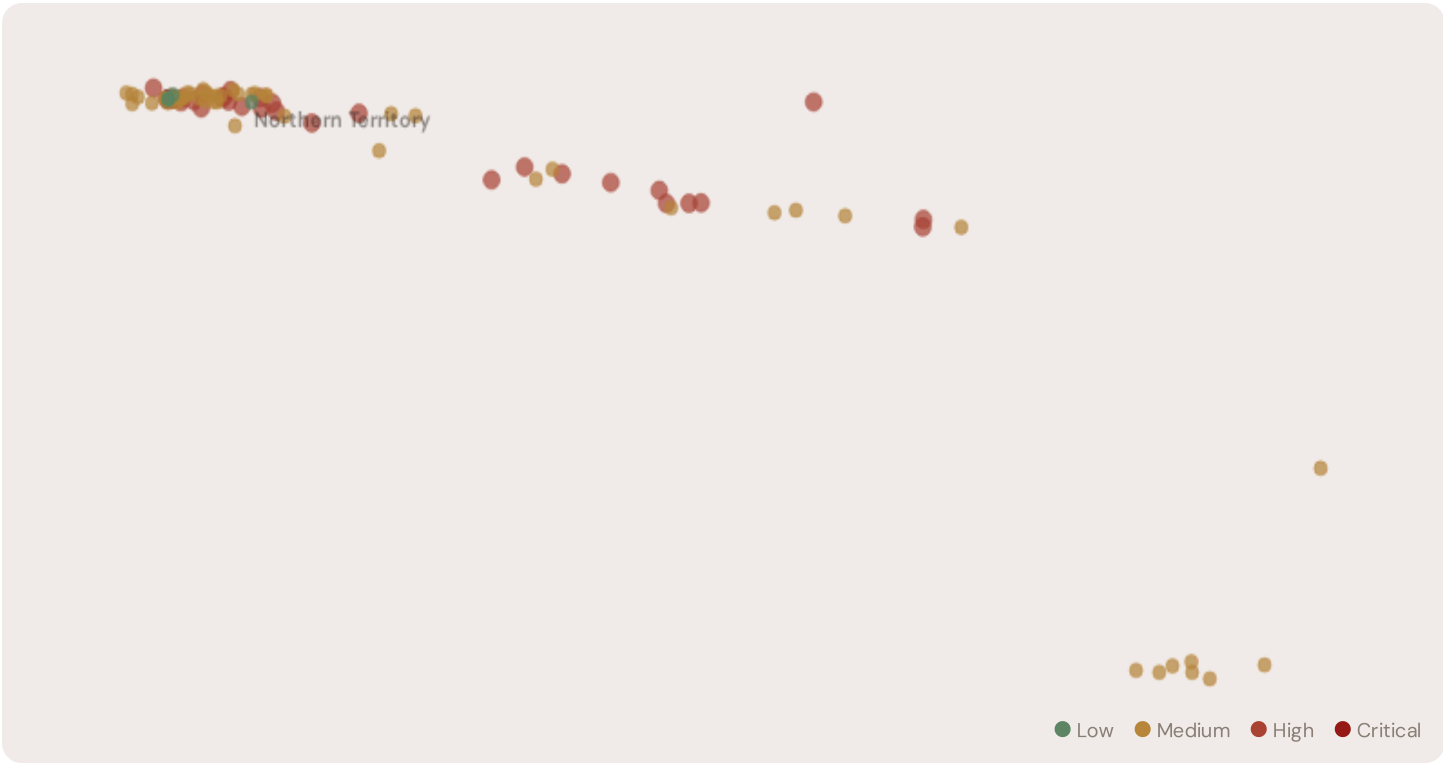
WORST SA4 REGION

Northern Territory

Avg R: 0.440 · 120 substations

The Northern Territory hosts **120 substations** across 1 region regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **43 substations (35.8%)** are classified High, with only **4** in the Low band. 63% of corridor substations score above the national fleet median of 0.383. The corridor median R of 0.436 reflects the local risk profile. The corridor concentrates the country's industrial heartland. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under national digitalisation timelines.

D.2 Corridor Map and Scoring



SUBSTATION	SA4 REGION	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Katherine SwStn 299	Northern Territory	0.723	High	0.114	1.022	0.701	1.538	0.750	2.326	V, I, E, S, T
Alice Springs DSub 275	Northern Territory	0.678	High	0.897	2.918	0.572	0.471	0.029	1.970	C, V, T
Darwin DSub 330	Northern Territory	0.677	High	0.621	1.660	0.235	0.704	1.481	0.671	C, V, E, S, T
Katherine SwStn 371	Northern Territory	0.663	High	0.168	2.614	0.017	0.203	1.513	2.611	V, S, T
Katherine TStn 373	Northern Territory	0.646	High	0.439	0.923	0.489	0.238	0.955	2.922	V, S, T
Katherine TStn 345	Northern Territory	0.637	High	0.693	2.340	0.510	1.945	0.487	2.054	C, V, E, T
Alice Springs ZSub 298	Northern Territory	0.618	High	0.567	0.600	0.632	1.755	0.732	2.095	I, E, S, T
Katherine BSP 357	Northern Territory	0.610	High	0.283	1.424	0.487	0.816	0.967	4.098	V, E, S, T
Darwin TStn 361	Northern Territory	0.609	High	0.007	2.302	0.495	1.809	0.528	4.027	V, E, T
Katherine TStn 377	Northern Territory	0.594	High	0.601	0.849	0.868	0.202	0.353	3.159	C, V, I, T

SUBSTATION	SA4 REGION	R_FINAL	BAND	C	V	I	E	S	T	ALERT
... 110 additional substations — explore full corridor in Digital Twin →										

D.3 Component Analysis

COMPONENT AVERAGES — NORTHERN TERRITORY VS FLEET			MODIFIER AVERAGES — NORTHERN TERRITORY VS FLEET		
C — Continuity	0.309 vs 0.273 (+13%)		R3 Consequence	0.956	fleet 0.949 ↓
I — Infrastructure	0.389 vs 0.328 (+19%)		R4 Graph Crit.	0.894	fleet 0.900 ↓
S — Saturation	0.475 vs 0.410 (+16%)		R6a Restoration	0.925	fleet 0.921 ↓
V — Voltage	1.009 vs 0.818 (+23%)		R6b Seismic	0.994	fleet 1.001 →
E — Economic	0.914 vs 0.824 (+11%)		R7 Digital Read.	1.008	fleet 1.010 →
T — Transition	1.785 vs 1.642 (+9%)				

The dominant risk driver across the Northern Territory corridor is **T (Transition)**, averaging 1.785 against a fleet average of 1.642 — occupying 3569% of its weight ceiling (w = 0.05). The second contributor is **V (Voltage)** at 1.009 vs fleet 0.818 (1009% of ceiling). Among modifiers, the R3 consequence multiplier averages 0.956 (fleet: 0.949), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 0.994, capturing the local seismic exposure — a dimension invisible to every competing framework.

D.4 SA4 Region Breakdown

REGION RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.

Northern Territory

120 subs · avg R = 0.440 · H:43 M:73

The region regions-level breakdown reveals significant intra-regional variation. **Northern Territory** leads with a mean R of 0.440 across 120 substations, while **Northern Territory** scores 0.440 — a spread of 0.000 within a single corridor. The SSI's substation-level granularity reveals that the Northern Territory is not uniformly stressed but contains distinct sub-corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

4 Northern Territory substations score $R \geq 0.650$, placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the Northern Territory sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

Asia-Pacific Context Card

How this works: Each edition includes one Asia-Pacific Context Card drawn from the \$16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the NSW–VIC corridor analysis hinges on Australia's continuity performance relative to regional peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Japan (003), CMIP6 climate hazard vs SE Asian peers (004), etc. The underlying data updates annually with AEMO/AER and national regulator publications.

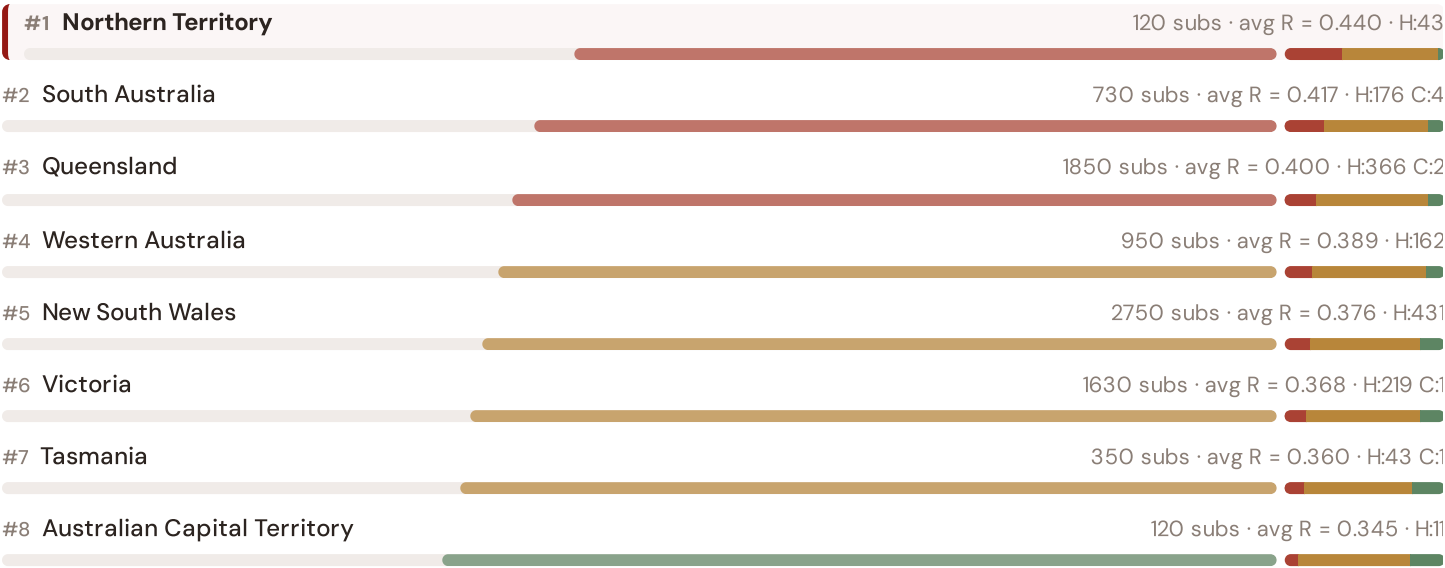


This month's key insight: The Northern Territory corridor deep-dive shows **75 substations** (of 120) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.309 is **1.1× the fleet average** of 0.273. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

State/Territory Risk Ranking — All 8 Australian States & Territories

Where does the NSW corridor sit within the national landscape? States sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.

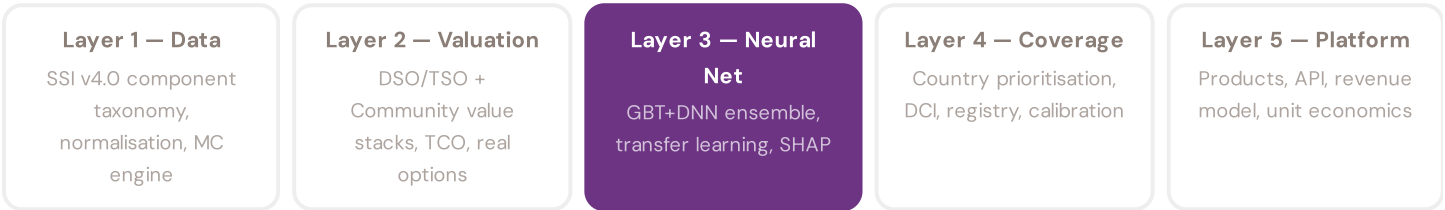


The Northern Territory ranks **#1 of 8 region regions** by mean R_median, placing it among the upper tier of national risk. The three most stressed are Northern Territory, South Australia, Queensland, while the three most resilient are Australian Capital Territory, Tasmania, Victoria. 4 of 8 score above the fleet average of 0.385. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the DNSP/TNSP and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.



LAYER 3 §0.5.6 · June 2026

Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history. Transparency converts model outputs into investment decisions.

KEY CONCEPTS

→ SHAP: game-theoretic feature attribution per prediction

→ Waterfall: fleet average → substation prediction

→ Direct traceability to SSI components and data sources

→ Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The training set comprises 8,500 substations with complete feature vectors. Average CI_width of 0.190 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 1122 Low, 5919 Medium, 1451 High, 8 Critical — ensures balanced training across all risk bands.

Coming next month:

LAYER 3

 Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 02

Mountain Region

Deep-dive into Region 10's grid risk profile — substation map, component analysis, region regions breakdown. European Context Card: Light-rural tier — limited substation density.

NEXT DATA REFRESH

Q3 2026 — o Quarterly Sources

none are due for refresh in Q3. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **TBD** (? variables →).

FLEET SPOTLIGHT

Northern Territory: Highest Risk Concentration

Northern Territory has the highest concentration of High/Critical substations: **43 of 120 (36%)** are in the upper risk bands — avg R = 0.440. This makes it the most acute risk corridor in the fleet.

Edition O2 will be published on the second Thursday of July 2026. Deep-dive region: **Region 10**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent an AEMO participant, DNSP, TNSP, state government, research institution, or industry body and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 003

Published 11 June 2026 · SSI Index v4.0.2 · 8,500 substations · ~9,200 grid lines · 6 components · 20 metrics · 6 modifiers
10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5